

Impact of Pharmacy Counseling on Compliance and Effectiveness of Combination Lipid-Lowering Therapy in Patients Undergoing Coronary Artery Revascularization: A Randomized, Controlled Trial

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This randomized, controlled trial evaluated the impact of personalized follow-up on compliance rates in high-risk patients receiving combination lipid-lowering therapy over 2 years. A random sample of 30 patients 7–30 days after cardiac surgery had baseline fasting low-density lipoprotein levels higher than 130 mg/dl. All patients received lovastatin 20 mg/day and colestipol 5 g twice/day. Weekly telephone contact was made with each patient for 12 weeks. Short- and long-term compliance was assessed by pill and packet counts and refill records. Compliance and lipid profile results were significantly better in the intervention group ($p < 0.05$) up to 2 years after the start of therapy than in the control group for all parameters except high-density lipoprotein. However, this effect was not apparent during the first 12 weeks of therapy. Short-term telephone follow-up favorably affected compliance and lipid profile results up to 2 years after start of therapy. (Pharmacotherapy 2000;20(4):410–416)

Reductions in low-density lipoprotein (LDL) cholesterol decrease the risk of recurrent myocardial infarction and death in healthy individuals and patients with coronary artery disease.^{1–7} Lowering LDL levels also slows the progression of coronary atherosclerosis.^{8, 9} In patients who have had saphenous vein coronary bypass graft surgery, aggressive lowering of LDL with a combination of lipid-lowering drugs (colestipol plus niacin or lovastatin) reduces atherosclerotic progression in grafts.^{10, 11}

Unfortunately, compliance with therapy in high-risk patients is poor. A retrospective

evaluation of outpatient medical charts indicated that of patients with coronary heart disease, only 35% were receiving lipid-lowering therapy,¹² and in those who had suffered a myocardial infarction the figure was just 25%.¹³ A survey of nearly 16,000 United States residents reported that 42% of hypercholesterolemic patients were aware of their condition, although only 4% were adequately treated and controlled.¹⁴ In the coronary drug project, nonadherence to lipid-lowering therapy was linked to a high rate of cardiac mortality.²

It is obvious that efforts to improve compliance and long-term use of lipid-lowering therapy are necessary, especially in high-risk patients. Studies have shown that lipid clinics managed by pharmacists produced improvements in these areas.^{15–17} A significantly greater number of patients enrolled in lipid clinics achieved National Cholesterol Education Program (NCEP) LDL treatment goals and adhered to therapy

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compared with patients receiving routine physician-based care. However, effective management requires long-term face-to-face counseling, which may not be feasible for large numbers of patients or for those living in outlying areas.

An alternative approach to enhance compliance and improve outcomes is telephone follow-up. Such contact has been successful in treatment of obesity and in maintenance of exercise regimens in middle-aged adults.¹⁸⁻²⁰ In addition, preprogrammed computerized telephone calls helped maintain dietary interventions in a group of hypercholesterolemic patients.²¹ To our knowledge, the benefit of telephone follow-up directed at enhancing compliance and outcomes with lipid-lowering therapy has not been evaluated. Therefore, we assessed the impact of personalized telephone follow-up on the rate of compliance in high-risk, hypercholesterolemic patients receiving combination drug therapy.

Methods

Patients

Enrollment took place over 7 months. Patients who had undergone coronary artery bypass graft (CABG) surgery or percutaneous transluminal coronary angioplasty (PTCA) in the previous 7-30 days were eligible. There were no restrictions with regard to age, race, or gender. Patients had to have a baseline fasting LDL above 130 mg/dl. They had to be able to read, understand, and speak English, and to have a telephone in their home. Written informed consent was obtained from each participant. Exclusion criteria were serum transaminase levels greater than 2 times the upper limit of normal; concomitant therapy with cyclosporine, warfarin, or erythromycin; and a history of significant gastrointestinal disease, including gastroesophageal reflux disease, peptic ulcer disease, Crohn's disease, and ulcerative colitis.

Design

Patients were recruited from the coronary care unit at St. Joseph Hospital. This facility is a university-affiliated tertiary care hospital in Omaha, Nebraska, and was licensed for approximately 400 beds at the time of the study. The Creighton University Cardiac Center, a freestanding outpatient clinic, is associated with the hospital. Because patients were located primarily in distant rural areas, it would have

been difficult to ensure routine face-to-face follow-up at the Cardiac Center or a lipid clinic.

While still hospitalized, all patients were prescribed lovastatin (Mevacor) 20 mg/day at bedtime and colestipol (Colestid) 5 g twice/day. Patients were responsible for filling and paying for these prescriptions. Each participant was extensively counseled on the appropriate use of the drugs at the time the prescriptions were written. All patients received dietary instruction before the start of drug therapy. Lipid profiles were measured at baseline, at 6 and 12 weeks after starting therapy, and at 1 and 2 years after enrollment. Patients were randomized to telephone contact or no telephone contact using a computer-generated list of random numbers.

Telephone Contact

A pharmacist telephoned patients at their home every week for 12 weeks. To ensure consistency in the information requested of the patients, the same pharmacist was involved in each patient contact and a standard set of questions was asked. Emphasis was placed on the importance of therapy in reducing the risk of recurrent cardiac events. Patients were questioned about when and where prescriptions were filled, how they paid for their prescriptions, potential side effects, overall well-being, and specific reasons for noncompliance when applicable.

Compliance

Compliance was determined by pill and packet counts (not performed within the patient's view) at the 6- and 12-week clinic visits. To assess long-term compliance, pharmacies at which patients filled their prescriptions were contacted at 1 and 2 years to document refill information. Patients returning more than 20% of prescribed pills at the 6- and 12-week visits, and those failing to fill 80% or more of their prescriptions at 1 and 2 years were considered noncompliant.

Statistical Analysis

Demographic and baseline clinical characteristics were summarized for each group as mean $\pm$ SD for continuous variables and as a percentage of the group for categorical or dichotomous variables. Univariate testing of independent clinical variables and lipid levels were compared between study and control groups using the χ^2 statistic for dichotomous variables and Student's *t* test for lipid parameters and other continuous variables.

Table 1. Clinical Characteristics and Demographics of Study Patients

Parameter	Telephone Contact	No Telephone Contact
Number of patients	15	15
Mean $\pm$ SD age (yrs)	64 $\pm$ 12	61 $\pm$ 12
Men/women	8/7	9/6
Race		
Caucasian	11	10
African-American	4	5
Mean $\pm$ SD yrs in school	11.9 $\pm$ 2.4	11.7 $\pm$ 2.9
PTCA	4	4
CABG	10	10
Both PTCA and CABG	1	1
Cardiovascular disease		
Myocardial infarction	6	6
Hypertension	5	7
Heart failure	3	2
Valvular disease	1	1
Noncardiovascular disease		
Diabetes mellitus	3	4
COPD, asthma	4	2
Thyroid disorder	0	1
Psychiatric illness	1	1
Arthritic disorder	2	2
Current smoker	2	1
Mean $\pm$ SD number of prescription drugs other than lipid-lowering therapy at study entry	5.2 $\pm$ 3.5	5 $\pm$ 4.2

COPD = chronic obstructive pulmonary disease.

The proportion of compliant patients between groups was evaluated using the χ^2 statistic. A p value less than 0.05 was considered statistically significant.

Results

Thirty patients were enrolled in the study, none of whom was lost to follow-up. Twenty had undergone CABG surgery, eight PTCA, and two

patients had both procedures in the preceding month. Fifteen patients were randomized to telephone contact and 15 to no telephone contact.

Baseline characteristics were similar for both groups (Table 1). No significant difference in compliance was seen for lovastatin tablets or colestipol packets between groups at 6 or 12 weeks (Table 2). In addition, the groups did not differ in the percentage of patients defined as being noncompliant (< 80% of doses consumed).

Changes in total cholesterol, LDL, high-density lipoprotein (HDL) cholesterol, and triglyceride levels were not significantly different between groups at 6 or 12 weeks (Table 3). After 12 weeks of treatment, 10 patients in each group had an LDL of 100 mg/dl or less.

Although this study had relatively low power due to the small sample (27% power with lovastatin, 33% power with colestipol at 1 yr), we detected significant changes in compliance as well as total cholesterol, LDL, and triglyceride levels. Long-term compliance based on prescription refill records indicated that at 1 and 2 years, patients receiving telephone contact demonstrated significantly better compliance than those not receiving this contact (Table 4). The number of patients defined as compliant ($\geq$ 80% of prescriptions filled) was also significantly greater in the telephone contact group. No patient found to be noncompliant at any time during the study reverted to compliant behavior.

All patients not receiving telephone follow-up were either compliant or noncompliant with both agents at 6 and 12 weeks. This was also true of the group receiving follow-up, with the exception of one patient who was compliant with both drugs at 6 weeks and noncompliant with both at 12 weeks.

Table 2. Short-Term Compliance by Study Patients

	Telephone Contact (n=15)		No Telephone Contact (n=15)	
	6 Weeks	12 Weeks	6 Weeks	12 Weeks
Compliance with lovastatin (%)	92	88	89	86
Compliance with colestipol (%)	93	90	90	88
No. of compliant patients ^a	13	12	12	12
Reasons for noncompliance				
Gastrointestinal side effects	1	1	1	1
Cost of prescriptions	0	0	1	1
Inconvenience	1	1	1	1
Lack of perceived benefit	0	1	0	0

Compliance was based on pill and packet counts.

^a80% or more of doses taken.

Table 3. Lipid Parameters in Study Patients at 6 and 12 Weeks after Start of Therapy Compared with Baseline (mean $\pm$ SD and % change)

	Telephone Contact	No Telephone Contact
Total cholesterol (mg/dl)		
Baseline	281 $\pm$ 62	289 $\pm$ 81
6 weeks	199 $\pm$ 57	225 $\pm$ 73
	$\downarrow$ 29.1 $\pm$ 10.3%	$\downarrow$ 22.3 $\pm$ 12.5%
12 weeks	201 $\pm$ 55	222 $\pm$ 75
	$\downarrow$ 28.5 $\pm$ 12.5%	$\downarrow$ 23.1 $\pm$ 17.4%
LDL (mg/dl)		
Baseline	192 $\pm$ 47	193 $\pm$ 51
6 weeks	128 $\pm$ 42	142 $\pm$ 47
	$\downarrow$ 33.4 $\pm$ 12.7%	$\downarrow$ 26.6 $\pm$ 14.7%
12 weeks	130 $\pm$ 43	140 $\pm$ 53
	$\downarrow$ 32.5 $\pm$ 13.2%	$\downarrow$ 27.3 $\pm$ 11.4%
HDL (mg/dl)		
Baseline	32.5 $\pm$ 5.3	33.7 $\pm$ 4.9
6 weeks	35.1 $\pm$ 6	36.6 $\pm$ 5.2
	$\uparrow$ 8.1 $\pm$ 8.9%	$\uparrow$ 8.7 $\pm$ 8.7%
12 weeks	35.4 $\pm$ 5.7	36.7 $\pm$ 4.9
	$\uparrow$ 8.8 $\pm$ 7.4%	$\uparrow$ 8.9 $\pm$ 3%
Triglycerides (mg/dl)		
Baseline	183 $\pm$ 51	194 $\pm$ 45
6 weeks	168 $\pm$ 70	182 $\pm$ 43
	$\downarrow$ 8.4 $\pm$ 3%	$\downarrow$ 6.4 $\pm$ 6.5%
12 weeks	166 $\pm$ 60	181 $\pm$ 47
	$\downarrow$ 9.2 $\pm$ 6.1%	$\downarrow$ 6.6 $\pm$ 8.6%

At the long-term follow-up evaluations, several patients either had changes in compliance from year 1 to year 2, or were compliant with lovastatin at both years, but not colestipol. No patient was compliant with colestipol but noncompliant with lovastatin. Three patients, one in the telephone contact group and two in the noncontact group, were compliant with lovastatin at both 1 and 2 years, but were noncompliant with colestipol. In each group, a single patient who was noncompliant with colestipol at both long-term evaluations was compliant with lovastatin at 1 year but noncompliant at 2 years. Finally, one patient in the group not receiving telephone follow-up was

compliant with both agents at 1 year, but became noncompliant with colestipol at 2 years.

Changes in total cholesterol, LDL, and triglyceride concentrations from baseline were substantially different between groups (Table 5), but changes in HDL were not. Eight patients in the telephone contact group had LDL levels of 100 mg/dl or less at 1 and 2 years compared with two in the no-contact group ($p < 0.05$).

Although only one subject each at 6 and 12 weeks stated that prescription cost contributed to noncompliance, it is possible that compliance rates were affected by insurance coverage for prescriptions. Thirty-three percent (5/15) of patients who received telephone follow-up had no insurance coverage for lipid-lowering therapy. In the group receiving no telephone contact, 27% (4/15) had no prescription coverage. Of noncompliant patients in the follow-up group, four of five receiving lovastatin and four of eight receiving colestipol would have had to pay for their prescriptions outright. In that group, 80% of patients who were responsible for the total cost of prescriptions were noncompliant with either drug. In the group that received no telephone contact, 100% of patients without supplemental prescription insurance coverage were noncompliant with both agents.

Other reasons for noncompliance obtained from patient interviews were gastrointestinal disturbances, perception that the regimen was inconvenient, and inability to detect benefit from therapy. We suspect that the higher rate of compliance with lovastatin compared with colestipol was related to the easier regimen, although this was not specifically addressed. There was no apparent difference in compliance based on sex; 10 women and 11 men were noncompliant at some point during the study.

Discussion

The results of our study indicate that weekly

Table 4. Long-term Compliance by Study Patients

	Telephone Contact		No Telephone Contact	
	1 Year	2 Years	1 Year	2 Years
Compliance with lovastatin (%)	71	63	47	39
Compliance with colestipol (%)	54	48	27	23
No. (%) of compliant patients				
Lovastatin	10 (67)	9 (60)	5 (33)	4 (27)
Colestipol	7 (47)	7 (47)	2 (13)	1 (7)

Compliance was based on prescription refill records.
 $p < 0.05$ for all values groups at 1 and 2 years.

Table 5. Lipid Values in Study Patients at 1 and 2 Years Compared with Baseline (mean $\pm$ SD and % change)

	Telephone Contact	No Telephone Contact	p Value
Total cholesterol (mg/dl)			
Baseline	281 $\pm$ 62	298 $\pm$ 81	
1 year	225 $\pm$ 49	260 $\pm$ 73	
	$\downarrow$ 19.9 $\pm$ 18.3%	$\downarrow$ 12.7 $\pm$ 22.5%	0.03
2 years	226 $\pm$ 60	259 $\pm$ 84	
	$\downarrow$ 19.5 $\pm$ 17.7%	$\downarrow$ 12.7 $\pm$ 23.8%	0.03
LDL (mg/dl)			
Baseline	192 $\pm$ 47	193 $\pm$ 51	
1 year	143 $\pm$ 40	163 $\pm$ 47	
	$\downarrow$ 25.7 $\pm$ 22.7%	$\downarrow$ 15.7 $\pm$ 21%	0.02
2 years	145 $\pm$ 53	164 $\pm$ 45	
	$\downarrow$ 24.3 $\pm$ 23%	$\downarrow$ 14.9 $\pm$ 20.6%	0.02
HDL (mg/dl)			
Baseline	32.5 $\pm$ 5.3	33.7 $\pm$ 4.9	
1 year	34.7 $\pm$ 6	35.7 $\pm$ 5.1	
	$\uparrow$ 6.7 $\pm$ 8.9%	$\uparrow$ 6 $\pm$ 9.3%	0.51
2 years	34.6 $\pm$ 5.8	35.6 $\pm$ 5.2	
	$\uparrow$ 6.5 $\pm$ 9.3%	$\uparrow$ 5.7 $\pm$ 6.2%	0.45
Triglycerides (mg/dl)			
Baseline	183 $\pm$ 51	194 $\pm$ 45	
1 year	163 $\pm$ 42	186 $\pm$ 39	
	$\downarrow$ 10.7 $\pm$ 14.5%	$\downarrow$ 4.3 $\pm$ 19.2%	0.04
2 years	164 $\pm$ 48	186 $\pm$ 43	
	$\downarrow$ 10.2 $\pm$ 13.7%	$\downarrow$ 3.9 $\pm$ 18.7%	0.04

telephone calls made for 12 weeks after the start of a combination of lipid-lowering drugs had no impact on compliance during that short time interval. This finding was surprising. One potential explanation is that initial adherence to the regimen in the nonintervention group was enhanced because early on those patients were acutely aware that their behavior was being assessed (Hawthorne effect). It is also possible that compliance rates were skewed toward the favorable side because of the relatively short time interval between a significant surgical event and the time of follow-up. Compliance rates may have been misleading because patients were highly aware of the existence and severity of their medical condition. Neither of the last two reasons is likely to have been a factor in the 1- and 2-year evaluations.

When compliance was assessed by prescription refill records over the ensuing 24 months, patients receiving telephone follow-up had greater compliance than those not receiving follow-up. Only 1 patient (7%) in the nonintervention group was compliant with both lovastatin and colestipol at 24 months, compared with 7 patients (47%) in the intervention group. Most important was that greater compliance

resulted in significant improvements in lipid values and in the number of patients achieving NCEP treatment goals. Data from the post-CABG lipid-lowering trial clearly showed the benefit of achieving NCEP targets in reducing the rate of atherosclerosis in vein grafts.¹¹

Our results are significant in light of overwhelming evidence that long-term compliance with lipid-lowering therapy is poor. Adherence to monotherapy is only 40% at 12 months, primarily because patients perceive that continued therapy is unnecessary.²² This indicates that positive reinforcement about the long-term benefits of treatment will affect compliance. Indeed, one of the primary messages given to patients in our study was the importance of continuing therapy for long-term benefits. Because patients take the drugs to alter essentially objective parameters, reminders of this need become that much more consequential.

Numerous theories and models for behavioral change exist, many of which contain components that may help explain some of the findings of this study. First, personal contact with a pharmacist, and more important, the same pharmacist, creates a helping relationship.²³ When patients were allowed and encouraged to express fears

and concerns to someone with whom they built a rapport, it helped establish a "therapeutic alliance" of sorts.²³ Positive feedback and support reinforce desired behaviors. In this way, a component of the process of change known as contingency management is fulfilled.²³ Follow-up by telephone also maintains a high level of patient awareness about information relevant to attitudes and behaviors that affect health-maintenance patterns (consciousness raising).²³ In conjunction with this, continued communication helps to disallow the possibility that a patient will maintain a state of denial regarding the potential gravity of an illness.²⁴

The Health Belief Model (HBM) hypothesizes that an individual's view that he or she is vulnerable to the consequences of a disease or illness, and the belief that these consequences will be severe, can predict how health-defining behaviors will be modified.²⁵ Drug therapy in our participants was begun just after a major surgical procedure. It is likely that patients accepted the surgeries as clear indicators of their vulnerability, and that consequences could be significant. Because of this, rates of adherence may have been more positively affected compared with rates one might expect to see with lipid-lowering therapy begun in the ambulatory setting based solely on laboratory analysis. The HBM also maintains that a patient will or will not be successful in changing behaviors depending on whether a belief is held that the intervention in question will positively affect health.²⁵ Reinforcement of the idea that adherence to the drug regimen was necessary and important to prevent further cardiac complications was a major component of the counseling provided by the pharmacist during this study. Also inherent to the HBM is the idea that a prompt that keeps an individual mindful of his or her beliefs about a health threat must be present.²⁵ The telephone calls served this function for our patients. However, in accordance with the HBM, one would suspect that because a patient's ideas and feelings about how vulnerable he or she is to a health threat may decrease as an acute event becomes more distant, willingness to continue with a drug regimen might wane.²⁶ Subjects in both groups had reduced rates of compliance at points of long-term evaluation compared with early in the study. However, those who received follow-up still maintained a higher overall rate of adherence to their drug regimen.

Compliance was enhanced when patients with chronic diseases considered their health care

provider to be accessible and inherently capable of meeting their needs according to the results of a study conducted to identify compliance predictors.²⁷ Although we did not formally assess patient satisfaction with follow-up by a pharmacist, it is possible that the subjects felt they were being shown more personal interest by a health care provider than they typically would, which may have augmented their desire to follow through with drug therapy.

Although lipid clinics can be effective in increasing compliance with therapy,¹⁵⁻¹⁷ they are not easily accessible to everyone. Their primary disadvantage is the requirement that patients return for face-to-face counseling. As a result, they are best suited for populations located in a metropolitan region. An alternative is telephone follow-up during which patients are encouraged to maintain compliance with therapy, laboratory tests, and physician visits. Telephone follow-up has favorably affected smoking cessation after myocardial infarction²⁸ and adherence to a low-cholesterol diet.²¹ Advantages inherent to this model of follow-up include a relatively lower cost than that associated with face-to-face counseling and implementation in populations (such as ours) that are geographically diverse. Our data indicate that telephone contact over 3 months after hospital discharge is better than no contact. It is unknown why this short-term intervention appears to have had such a lasting effect. Further study is necessary to determine if this observation is consistently demonstrable, and how long the residual effects remain.

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